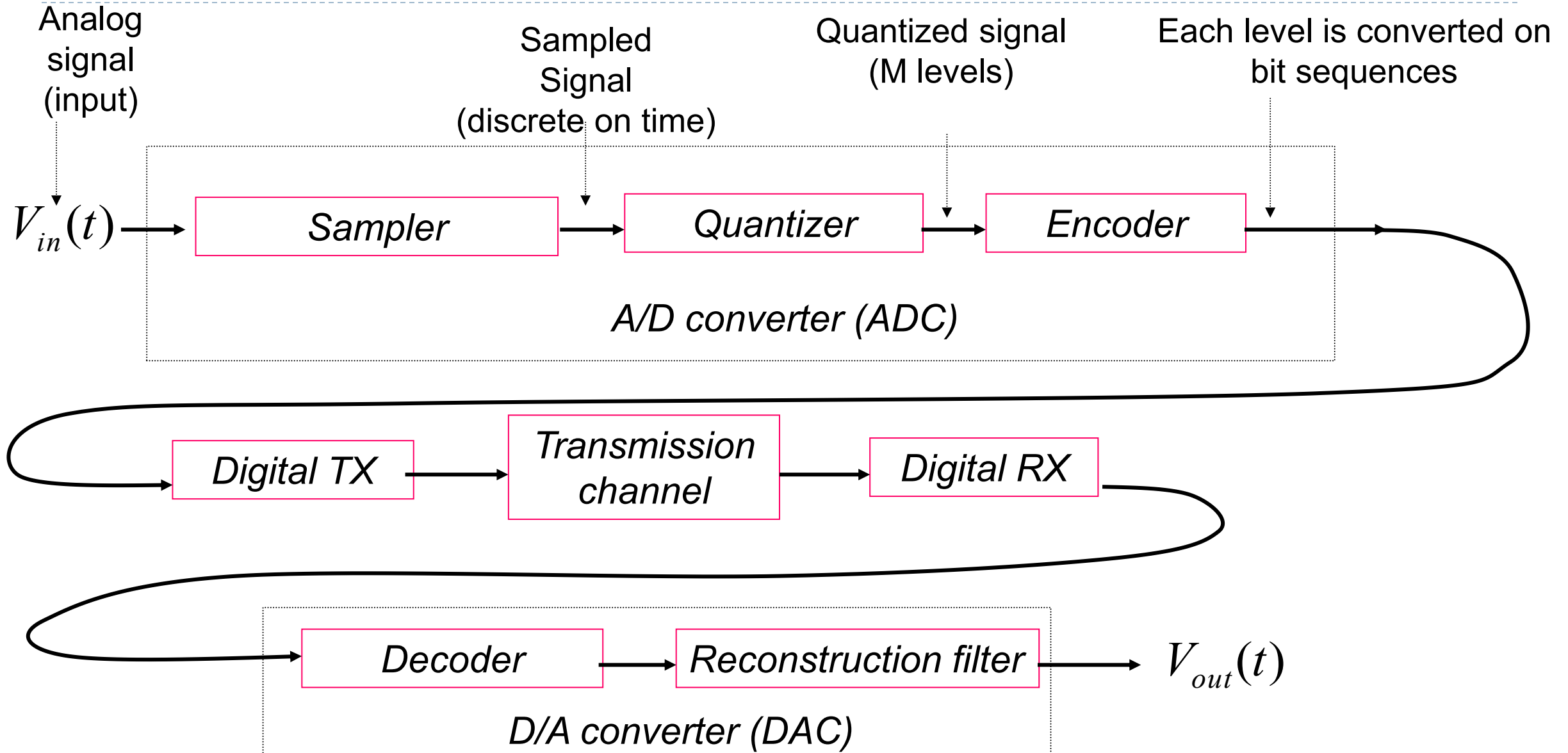
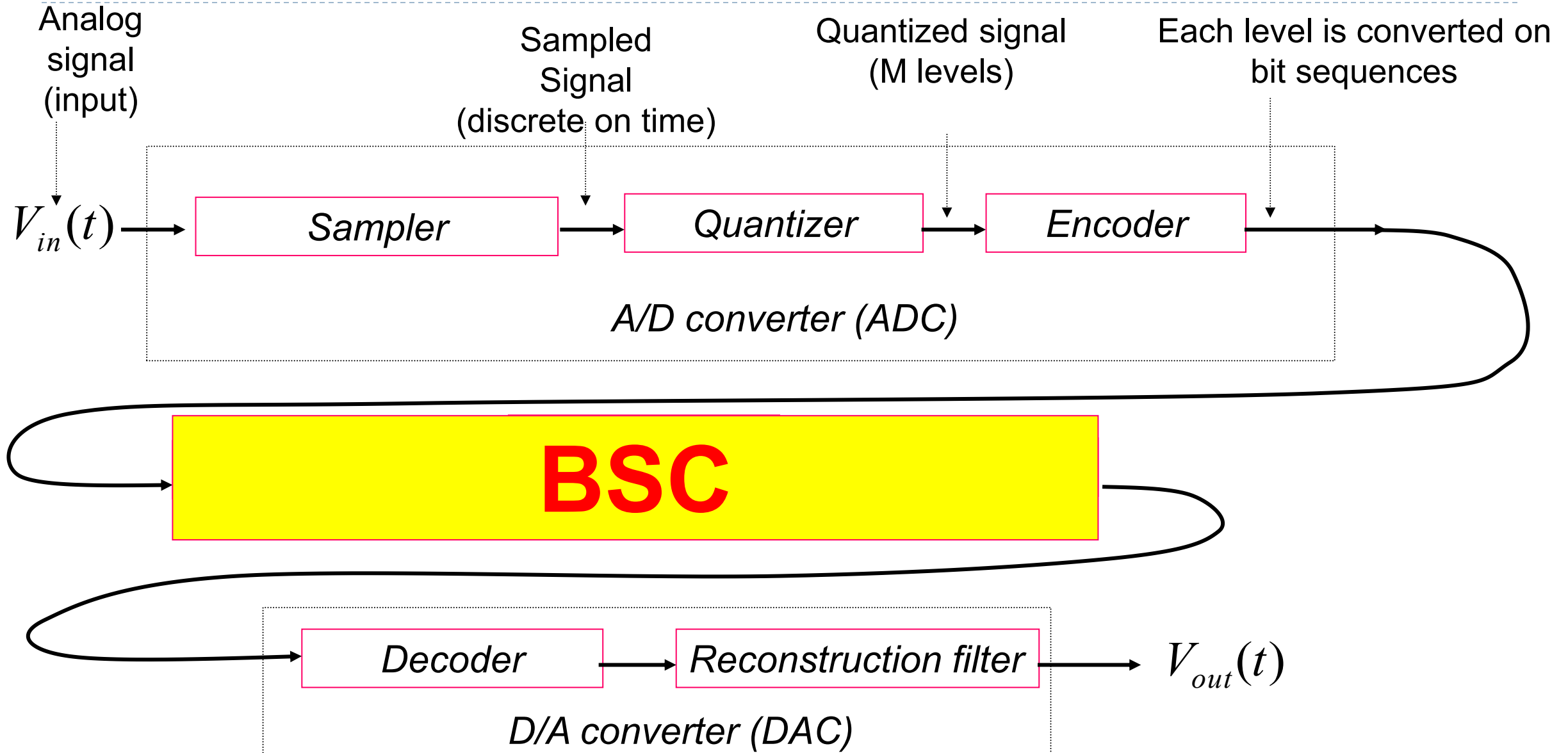


TLC Virtual LAB#2

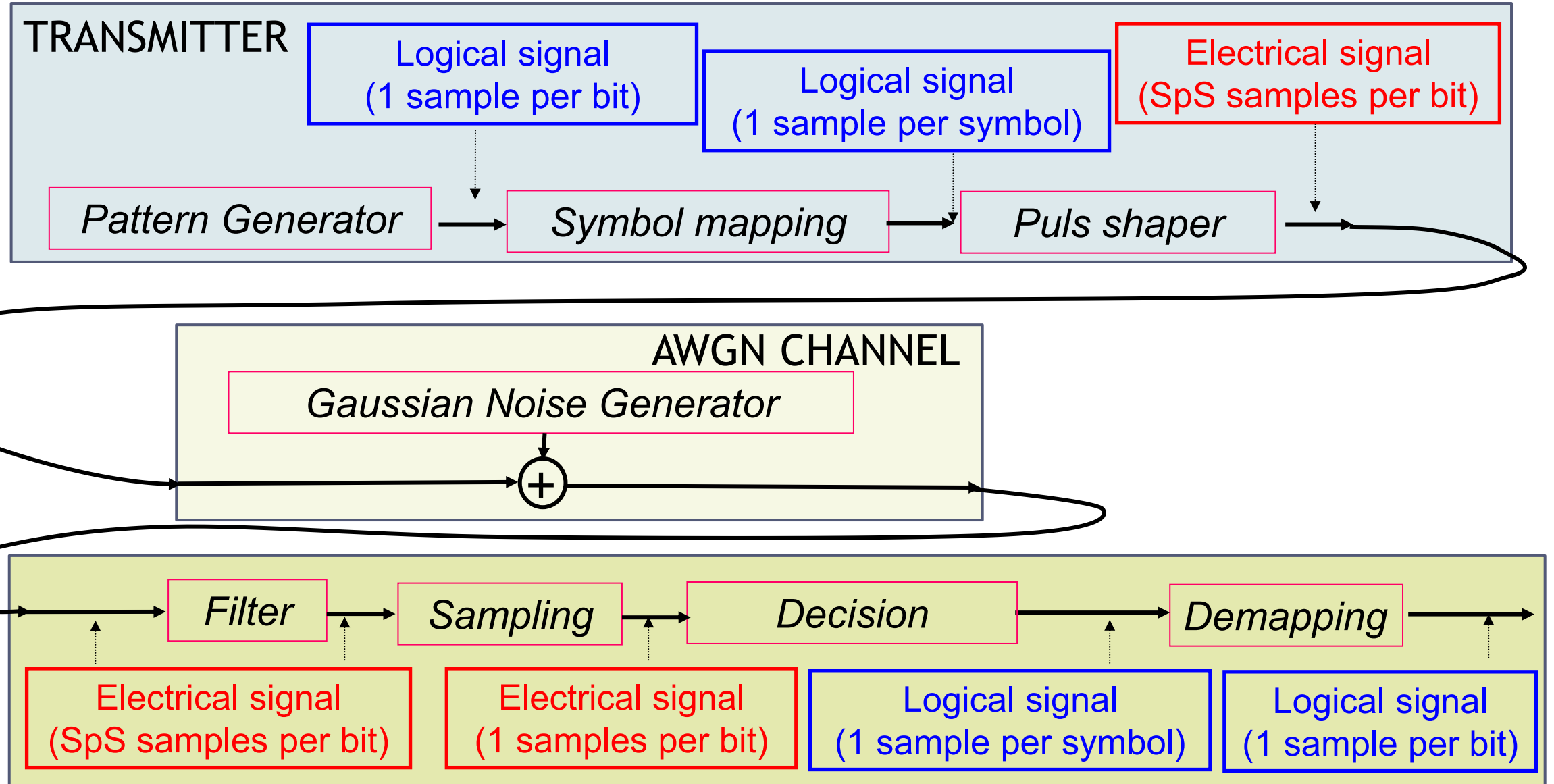
A TRANSMISSION SYSTEM



A TRANSMISSION SYSTEM



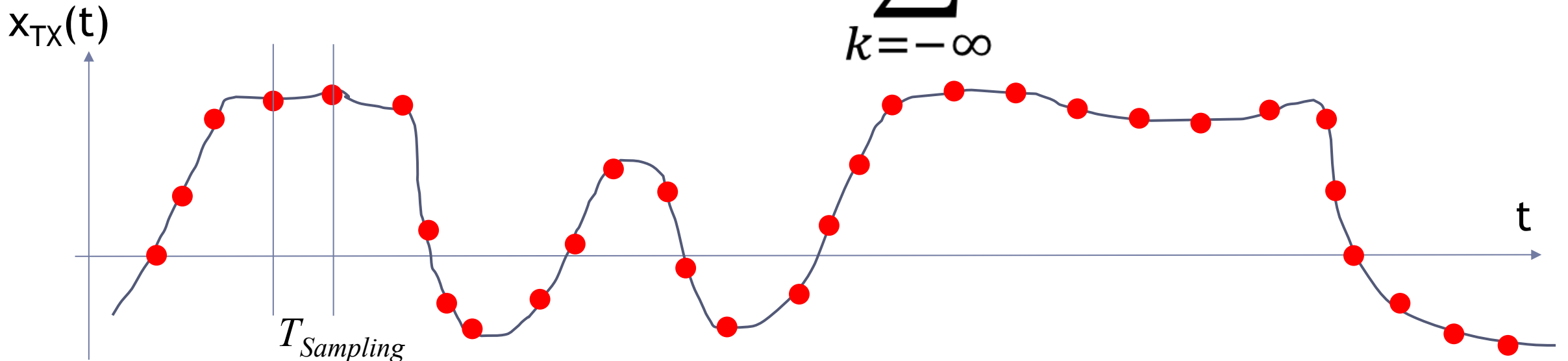
THE DIGITAL TRANSMISSION SYSTEM



TIME DOMAIN SIMULATION

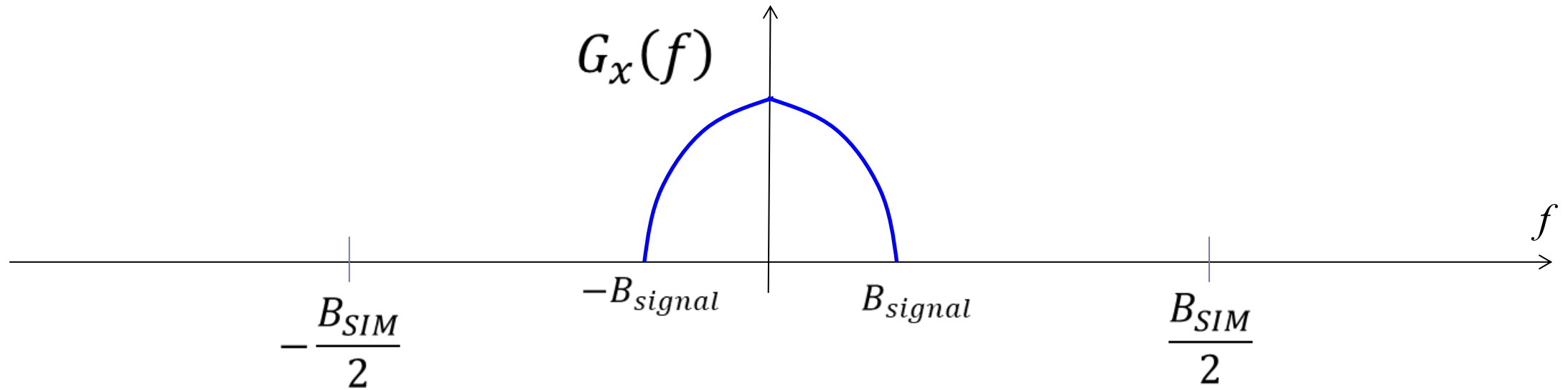
- ▶ We want to carry out a time domain simulation of a digital transmission system
- ▶ A numerical simulator can only handle sampled version of signals

$$x_{TX} = \sum_{k=-\infty}^{+\infty} \alpha_k s(t - kT_{Symbol})$$



CHOICE OF SIMULATION BANDWIDTH

- ▶ Given a sampling interval T_{Sampling} , the simulation bandwidth is equal to $[f : -B_{\text{SIM}}/2 \leq f \leq B_{\text{SIM}}/2]$, for $B_{\text{SIM}} = 1/T_{\text{Sampling}}$



- ▶ Depending on the signal we need to simulate, a proper choice of B_{SIM} must be done
- ▶ In order to guarantee a good precision of the considered digital models, some guard-band must be considered

$$B_{\text{sim}} \gg B_{\text{signal}}$$

CHOICE OF SIMULATION BANDWIDTH

- ▶ When simulating signal carrying digital data, a major time frame is the symbol duration T_{Symbol}
- ▶ B_{signal} depends on T_{Symbol}
- ▶ A typical choice is to have an integer number of samples in T_s so that

$$T_{Sampling} = \frac{T_{Symbol}}{SpS}$$

where SpS is the number of samples per symbol

HOW MANY SAMPLES PER SYMBOL?

- ▶ How many samples per bit?
 - ▶ It depends on the bandwidth you want to simulate...

$$B_{SIM} = \frac{1}{T_{Sampling}} = \frac{SpS}{T_{Symbol}} = SpS \cdot R_S$$

NORMALIZING FREQUENCY AND TIME

- ▶ To implement a very general tool, you do not need to specify symbol rate
- ▶ Theory findings are bit-rate/symbol-rate agnostic

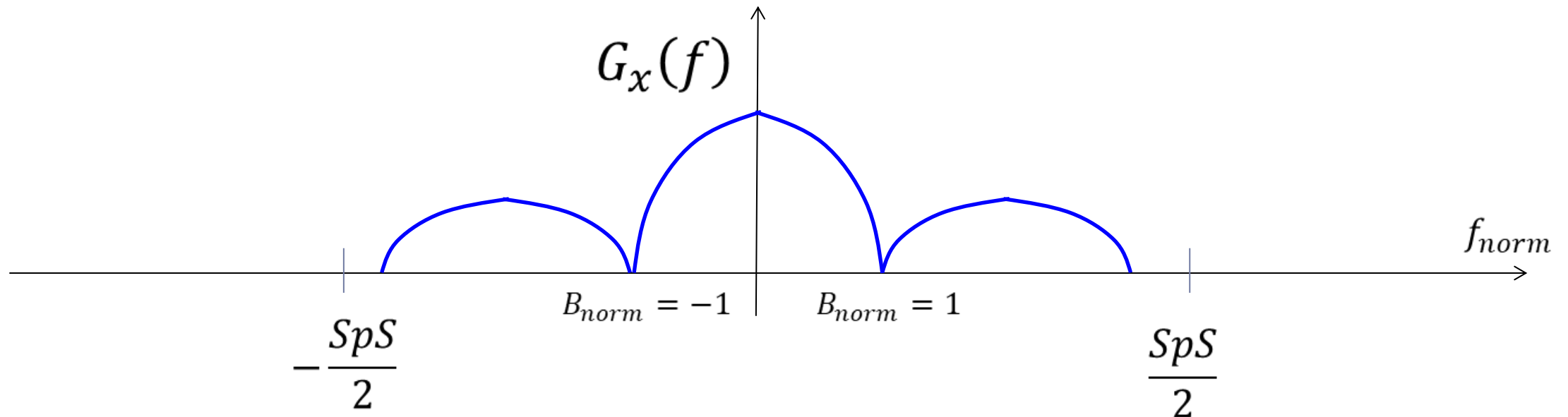
$$P_b(e) = f\left(\frac{E_b}{N_0}\right)$$

- ▶ Consider normalized frequency and time

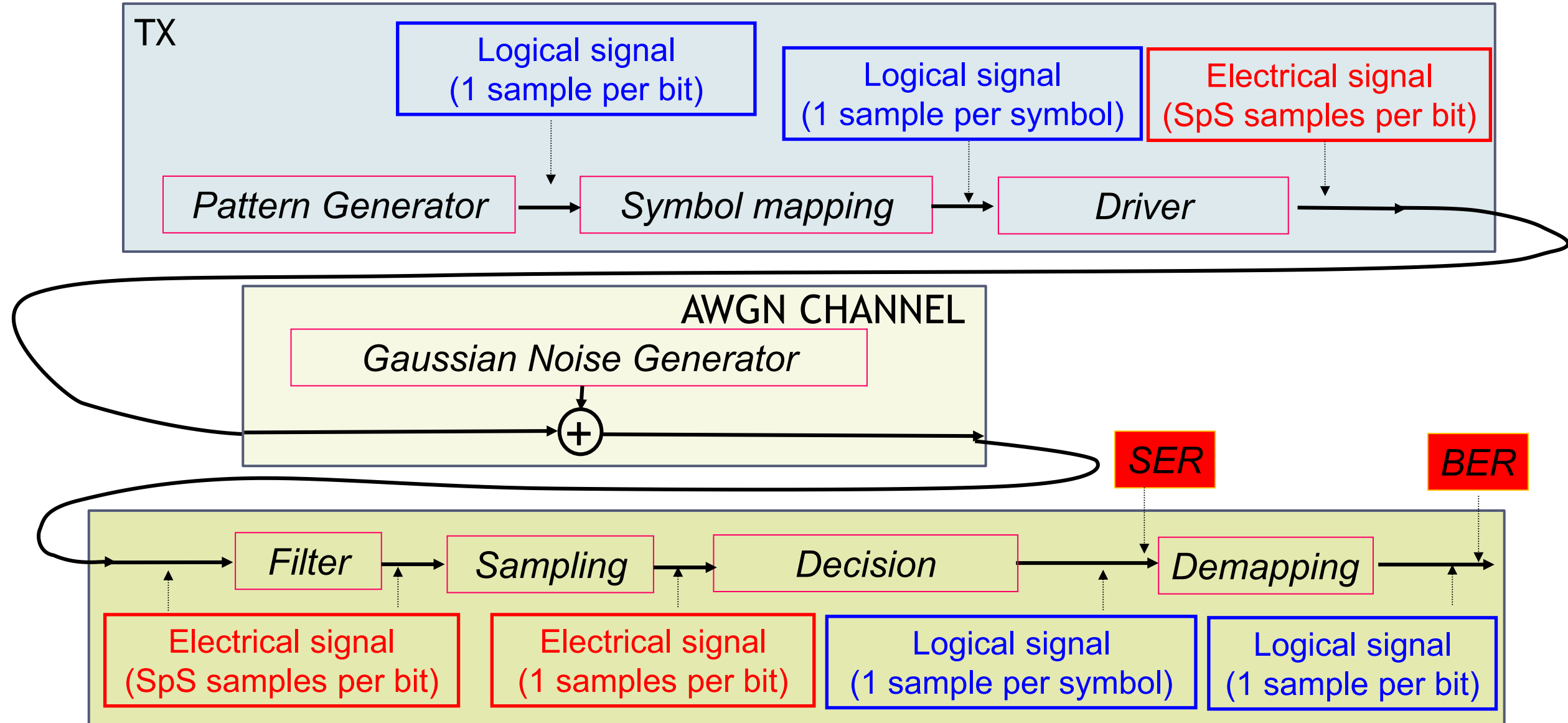
$$f_{norm} = \frac{f}{R_S} \quad t_{norm} = \frac{t}{T_{Symbol}}$$

NORMALIZED FREQUENCY

- ▶ The simulation bandwidth is SpS
- ▶ A bandwidth equal to 1, is equivalent to R_S



PAM-M block diagram: SER/BER evaluation



TRANSMITTER - I

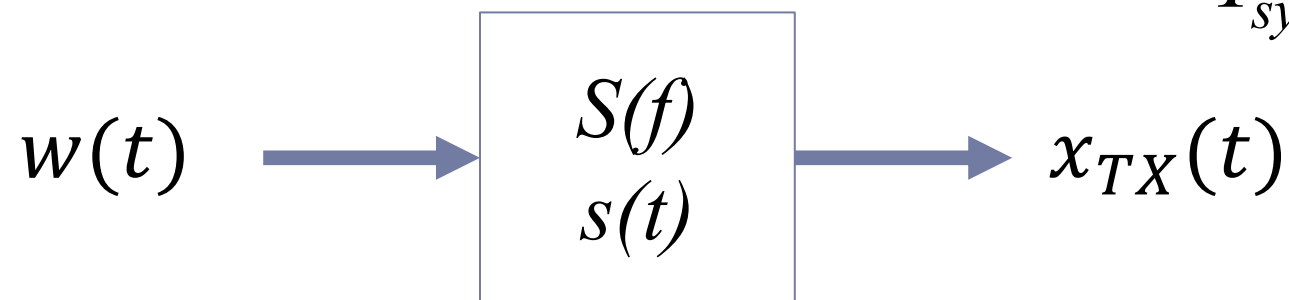
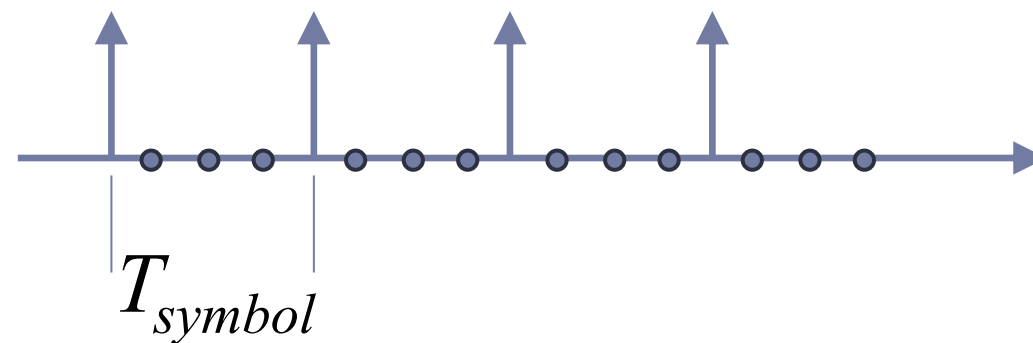
- ▶ Pattern generator
 - ▶ Random bit generation using Matlab integer uniform distributed pseudo-random generator or PN sequence
- ▶ **randi**
- ▶ **comm.PNSequence**

TRANSMITTER - II

- ▶ Symbol mapping
 - ▶ Group BpS bits in a symbols
 - ▶ Based on the labeling, select a symbol
 - ▶ Transmit the selected α_k
- ▶ **pammod**
- ▶ **comm.PAMModulator**

TRANSMITTER - IV

$$w(t) = \sum_{k=-\infty}^{\infty} \alpha_k \delta(t - kT_{symbol})$$



$$s(t) = w(t) * s(t)$$

$$x_{TX}(t) = \sum_{k=-\infty}^{\infty} \alpha_k s(t - kT_{symbol})$$

TRANSMITTER - III

- ▶ Pulse shaping
 - ▶ Synthesize a digital filter to apply the selected pulse shape
 - ▶ **filter**
 - ▶ **dsp.FIRFilter**
 - ▶ **comm.RaisedCosineTransmitFilter**

NRZ – PAM-2

$$x_{TX} = \sum_{k=-\infty}^{+\infty} \alpha_k s(t - kT_{Symbol})$$

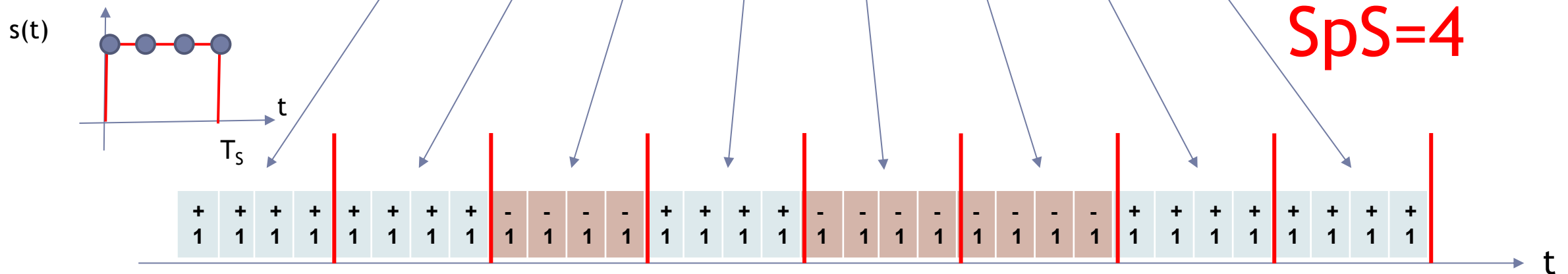
BIT

1	1	0	1	0	0	1	1
---	---	---	---	---	---	---	---

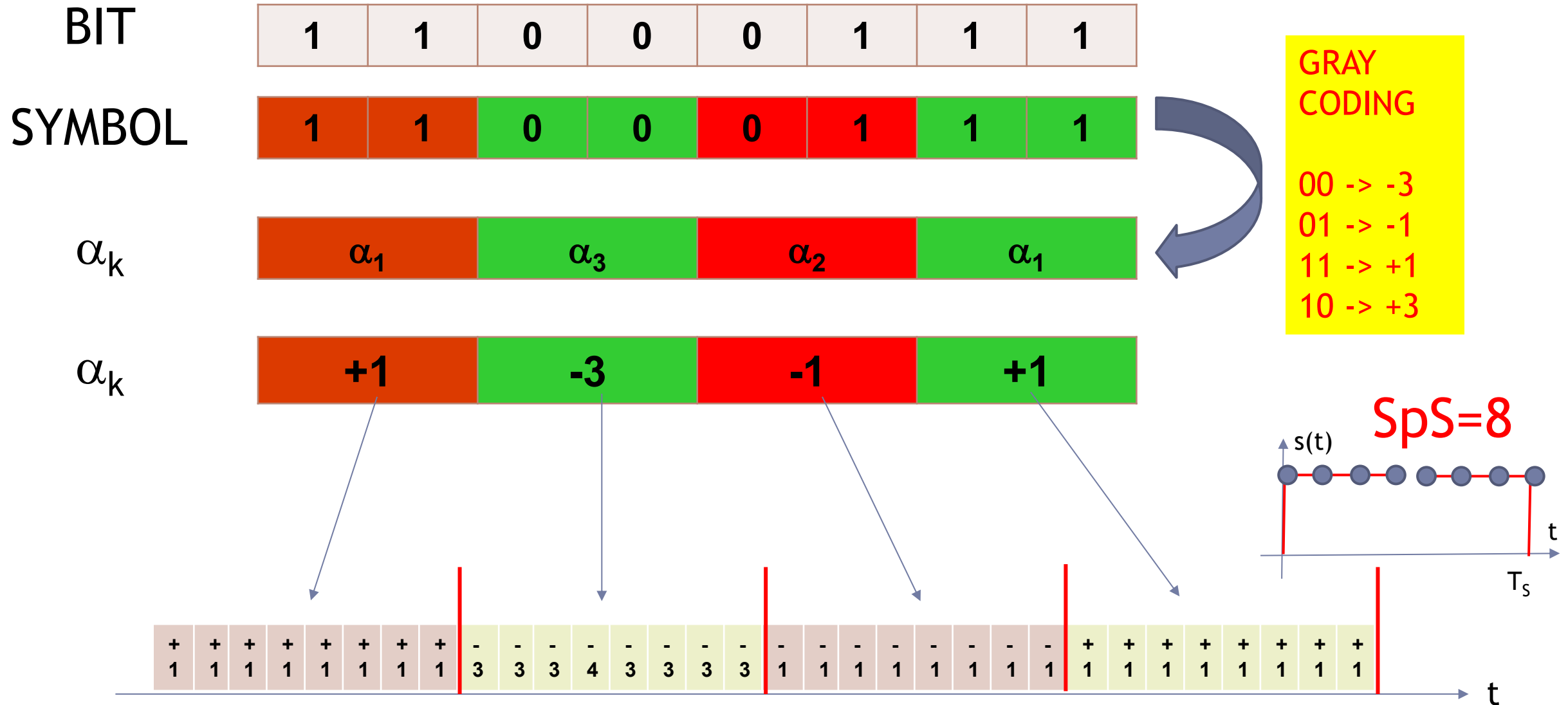
Example: antipodal

ALFA

+1	+1	-1	+1	-1	-1	+1	+1
----	----	----	----	----	----	----	----



NRZ – PAM-4



AWGN CHANNEL

- ▶ Noise generator: white and Gaussian

randn

- ▶ How to evaluate N_0 ?

$$\sigma_n^2 = \frac{N_0}{2} B_{SIM}$$

- ▶ How to evaluate N_0 ?

$$\frac{E_b}{N_0} = \frac{P_S \cdot T_b}{N_0} \quad \Rightarrow \quad N_0 = \frac{P_S}{\left(\frac{E_b}{N_0}\right) R_b}$$

RECEIVER - I

- ▶ Filter: you can reuse shaping filters!

RECEIVER - II

- ▶ Sampling

- ▶ Determine the optimum sampling instant

`samples_rx=x_RX[pos:SpS:end];`

- ▶ Decision

- ▶ Compare detected samples with threshold/thresholds
 - ▶ Which is/are the optimal threshold/s?

For PAM-2

`bits_rx=samples_rx>threshold;`

RECEIVER - III

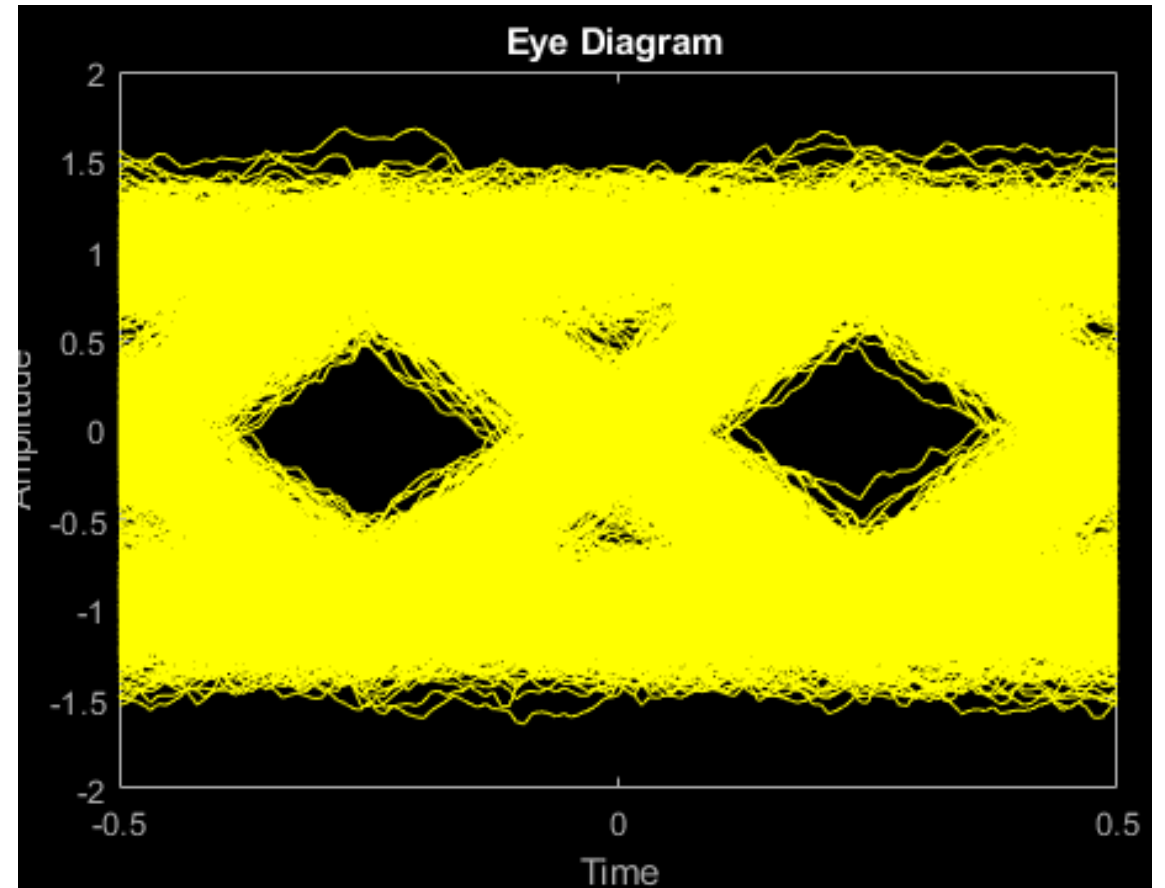
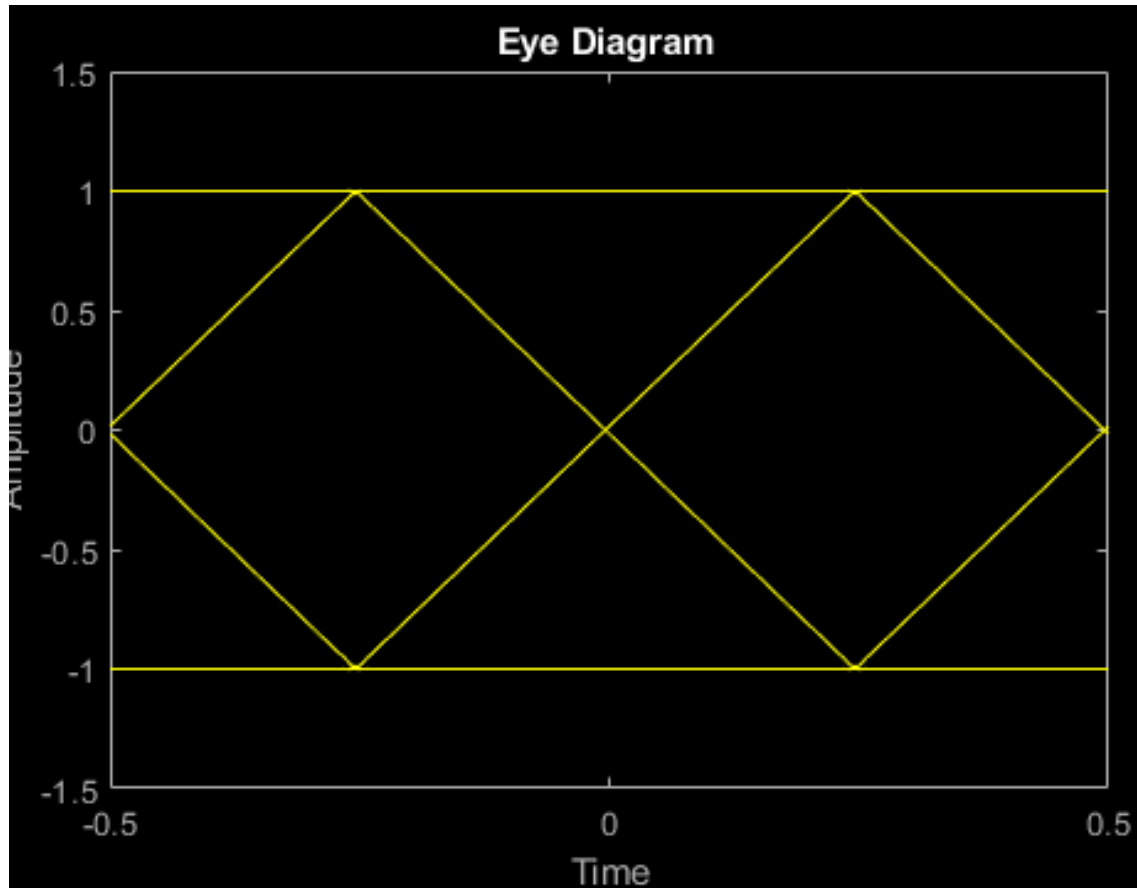
- ▶ Error counting
 - ▶ Compare TX and RX bit sequences, counting errors

biterr

comm.ErrorRate

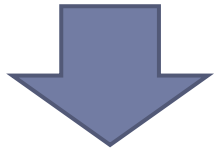
EYE DIAGRAM

eyediagram

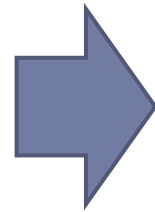


HOW MANY BITS SHOULD I SIMULATE?

$$BER = \frac{\#errors}{\#bits}$$



$$\#errors = \#bits \cdot BER$$



$$\#bits = \frac{\#errors}{BER}$$

- ▶ As we are “counting errors”, they should be “enough” for an accurate BER estimation
 - ▶ You must count at least 10 errors, possibly 100